

DW_lp_piped_div

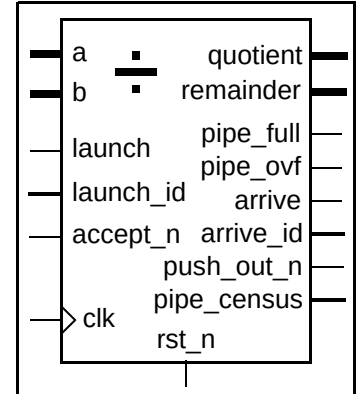
Low Power Pipelined Divider

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Built-in pipelining and power management
- Automatically enables register retiming
- Parameterized operand widths
- Parameterized pipeline stages
- Launch identifier tracking propagation
- Operand isolation capability on a and b

Revision History



Description

DW_lp_piped_div performs a pipelined division based on two operands with the added benefit of power savings. Pipeline control is integrated and applied to pipelined register levels (when configured in) to minimize power consumption. Pipeline register re-timing is automatically enabled for balancing between logic stages.

Component pins are described in [Table 1-1](#) and configuration parameters are described in [Table 1-2](#).

Table 1-1 Pin Descriptions

Pin Name	Width	Direction	Function
clk	1 bit	Input	Input clock
rst_n	1 bit	Input	Asynchronous or synchronous reset depending on rst_mode parameter (active low)
a	<i>a_width</i> bits	Input	Dividend
b	<i>b_width</i> bits	Input	Divisor
quotient	<i>a_width</i> bits	Output	Quotient of a divided by b
remainder	<i>b_width</i> bits	Output	Remainder of a divided by b.
launch	1 bit	Input	Control to begin a new divide operation
launch_id	<i>id_width</i> bits	Input	Identifier for the corresponding asserted launch
pipe_full	1 bit	Output	Upstream notification that pipeline is full
pipe_ovf	1 bit	Output	Status flag indicating pipe overflow
accept_n	1 bit	Input	quotient result accepted from downstream logic (active low)

Table 1-1 Pin Descriptions (Continued)

Pin Name	Width	Direction	Function
arrive	1 bit	Output	quotient result is valid
arrive_id	id_width bits	Output	launch_id from the originating launch that produced the quotient result
push_out_n	1 bit	Output	Push performed to downstream FIFO element (active low)
pipe_census	M bits	Output	Number of pipeline register levels currently occupied Note: The value of M is equal to the larger of 1 or $\text{ceil}(\log_2(\text{in_reg} + \text{stages} + \text{out_reg}))$. Example: if $\text{in_reg} = 1$, $\text{stages} = 2$, $\text{out_reg} = 1$, then $M = 2$

Table 1-2 Parameter Description

Parameter	Values	Description
a_width	≥ 1 Default: 8	Word length of a
b_width	≥ 1 Default: 8	Word length of b
id_width	1 to 1024 Default: 8	Width of launch_id
in_reg	0 or 1 Default: 0	Input register control 0: No input register 1: Include input register
stages	1 to 1022 Default: 4	Number of pipeline stages
out_reg	0 or 1 Default: 0	Output register control 0: No output register 1: Include output register
tc_mode	0 or 1 Default: 0	Two's complement control 0: Unsigned 1: Signed
rst_mode	0 to 1 Default: 0	Reset mode 0: Asynchronous reset 1: Synchronous reset
rem_mode	0 or 1 Default: 0	Remainder output control 0: $a \bmod b$ (remainder has sign of divisor) 1: $a \text{ rem } b$, a / b (remainder has sign of dividend)

Table 1-2 Parameter Description (Continued)

Parameter	Values	Description
op_iso_mode	0 to 4 Default: 0	Operand isolation mode (controls datapath gating for minPower flow) Allows you to set the style of minPower datapath gating for this module ■ 0: Use the DW_lp_op_iso_mode^a synthesis variable ■ 1: 'none' ■ 2: 'and' ■ 3: 'or' ■ 4: Preferred gating style: 'and' Datapath gating is inserted only when there are no input registers on the operands at the component boundary. When inserted, datapath gating circuits are placed immediately after the input ports of the component (see Figure 1-2 on page 5).

a. The DW_lp_op_iso_mode synthesis variable is available only in Design Compiler.

DW_lp_op_iso_mode sets a global style of datapath gating. To use the global style, set *op_iso_mode* to '0'. Note that If the *op_iso_mode* parameter is set to '0' and DW_lp_op_iso_mode is either not set or set to 0', then no datapath gating is inserted for this component.

Table 1-3 Synthesis Implementations

Implementation Name	Implementation	License Feature Required
rtl	Synthesis model	■ DesignWare (P-2019.03 and later) ■ DesignWare-LP^a (before P-2019.03)

a. For Design Compiler versions before P-2019.03, see [“Enabling minPower”](#) on page 12.

Table 1-4 Simulation Models

Model	Function
DW03.DW_LP_PIPED_DIV_CFG_SIM	Design unit name for VHDL simulation
dw/dw03/src/DW_lp_piped_div_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_lp_piped_div.v	Verilog simulation model source code

Functional Description

Block Diagram

Figure 1-1 shows the block diagram of the DW_lp_piped_div component:

Figure 1-1 DW_lp_piped_div Basic Block Diagram

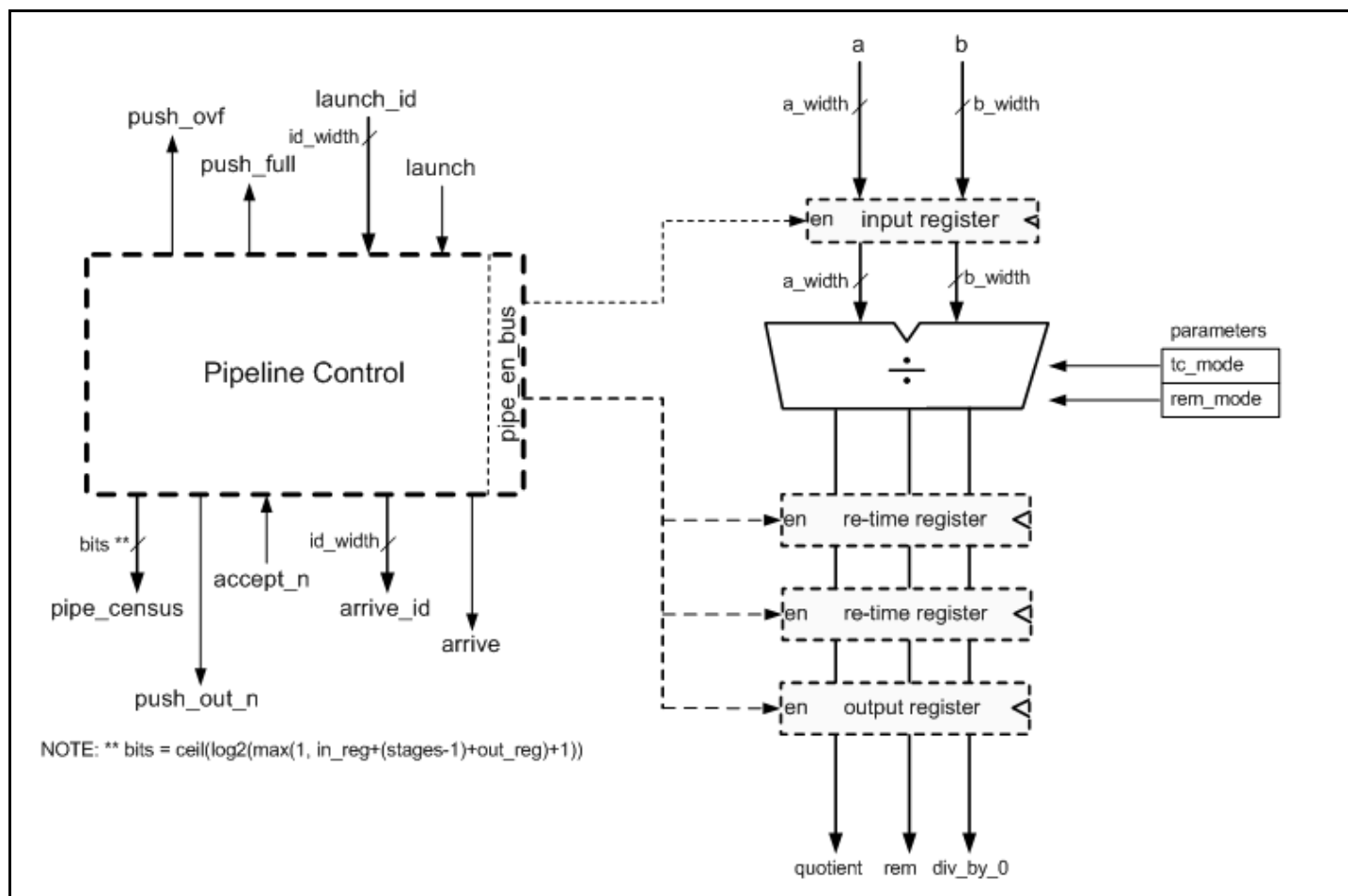
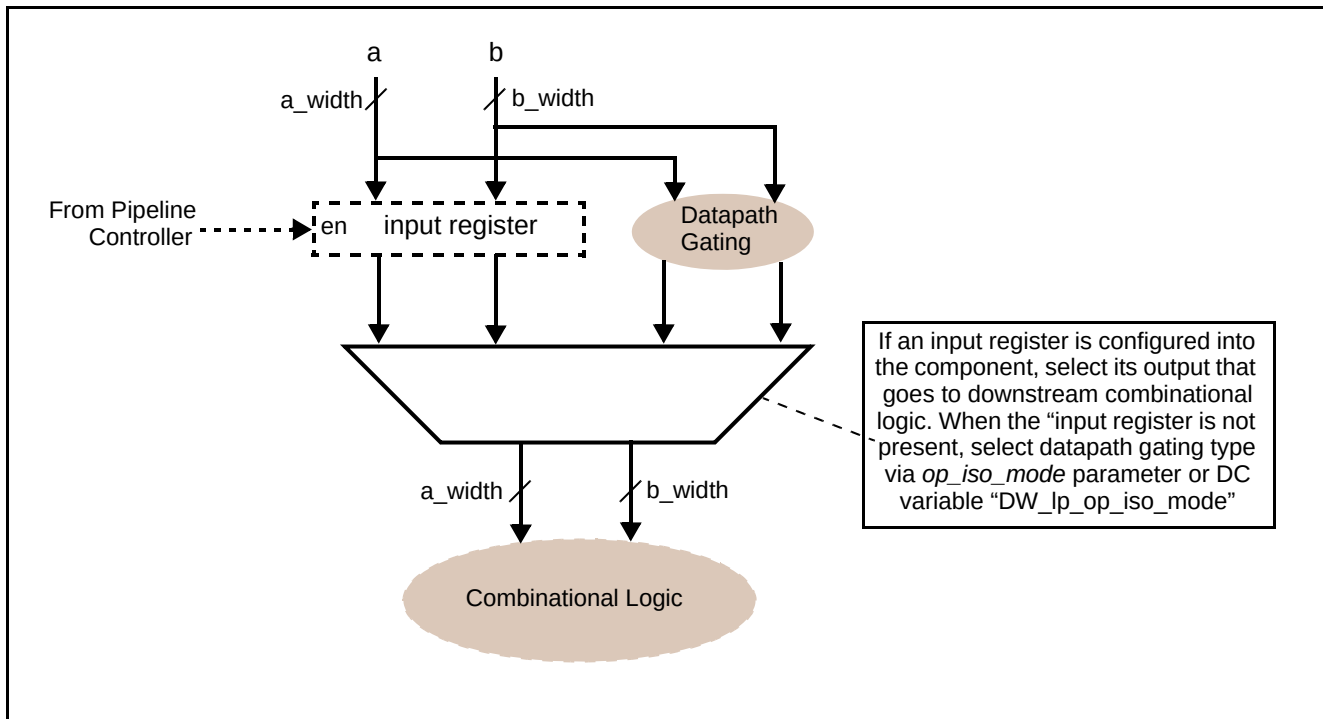


Figure 1-2 shows where datapath gating is inserted when the *op_iso_mode* parameter enables it.

Figure 1-2 Location of Datapath Gating (If Inserted)



Pipelining

The DW_lp_piped_div is configurable to embed pipeline register levels. Setting the value for the parameters *in_reg*, *stages*, and *out_reg* (see Table 1-5 on page 6) determines the number of pipeline register level(s) that are inserted. Therefore, depending on the parameter *in_reg*, *stages*, and *out_reg* settings, the number of clock cycles for the quotient result to propagate varies. The *rem_mode* parameter determines the behavior of the output.

This DW_lp_piped_div is designed to make it easy to pipeline the divider using the register retiming features of Design Compiler (DC). It also contains parameter controlled input and output registers which will stay in place at their respective block boundary, that is, they are not allowed to be moved by the DC register retiming feature.

The parameter *stages* refers to the number of logic stages desired after register retiming is performed. The number of register levels is not necessarily the same as the number of logic stages. If no input or output registers are used (*in_reg* = 0 or *out_reg* = 0), then there is one fewer register level than logic *stages*. If either an input register or output register is specified, then the number of register levels is the same as the number of logic *stages*. If both input and output registers are specified, then the number of register levels is the number of logic *stages* + 1. Refer to Table 1-5 which describes this. The number of pipeline register levels that can be retimed is always *stages* - 1.

Table 1-5 Number of Pipeline Register Levels

in_reg	out_reg	Number of Pipeline Register Levels
0	0	stages - 1
0	1	stages
1	0	stages
1	1	stages + 1

Pipeline Control and Power Savings

Running in parallel to the pipeline register levels is pipeline control logic (as seen in [Figure 1-1](#)) that monitors the activity. In cases where there is inactivity on a particular register level of the pipeline, the pipeline control disables those levels to promote power savings. Furthermore, if using the Synopsys Power Compiler tool, the presence of the pipeline control and its wiring to the pipeline register levels provides an opportunity for increased power reduction in the form of clock gating.

Along with the potential power savings that the pipeline control provides, it can be utilized to improve performance in cases where intermittent launch operations are present and there contains FIFO structures upstream and downstream of the DW_lp_piped_div. The handshake is made between the DW_lp_piped_div and the external FIFOs via the `accept_n` and `pipe_full` ports. Effectively, the DW_lp_piped_div can be considered part of the external FIFO structures. The performance gain comes when inactive (bubbles) stages are detected. These pipeline 'bubbles' are removed to produce a contiguous set of active pipeline stages. The result is empty pipeline slots at the head of (or entering) the DW_lp_piped_div pipeline for new operations to be launched. Advancing the shifting of operations through the pipeline when a valid quotient result is available (`arrive = 1`) is controlled by the `accept_n` input. When the divider pipeline is full of active entries, the `pipe_full` output is '1'. To disable this feature in cases where no external FIFOs are present, set the `accept_n` input to '0' which will effectively eliminate any flow control. At the same time, the `pipe_full` output would always be '0'.

To assist in tracking of launched operands, the pipeline control logic provides interface ports called `launch_id` and `arrive_id`. The `launch_id` input is assigned a value during an active launch operation. Given that `launch_id` values are unique in successive launch operations, the quotient results can be distinguished from one another with the assertion of `arrive` and the associated `arrive_id`. The `arrive_id` is the `launch_id` from the originating launch that produced the valid quotient result.

No Pipeline Register Levels Specified

In cases where no pipelining is required through the DW_lp_piped_div (`in_reg = 0`, `stages = 1`, and `out_reg = 0`), the pipeline control flow control handshaking/status signals still remain active and meaningful with one exception. The `pipe_census`, which is intended to count the number of active pipeline register levels, becomes irrelevant and is fixed to '0'; see [Figure 1-5](#) on page 10.

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If values other than 1 or 0 are present on a clock port, the following message is displayed:

```
WARNING: <instance_path>.<clock_name>_monitor:  
at time = <timestamp>, Detected unknown value, x, on <clock_name> input.
```

To suppress only this warning message for all DWBB components, use the following macro:

- Define the DW_DISABLE_CLK_MONITOR macro. You can define this macro in the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_DISABLE_CLK_MONITOR
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_DISABLE_CLK_MONITOR (which is used for the Synopsys VCS simulator)
```

This message is also suppressed using the DW_SUPPRESS_WARN macro explained earlier.

- If a divide-by-zero operation is attempted, the following message is displayed:

```
WARNING: <instance_path>:  
at time = <timestamp>: Division by zero.
```

To suppress this message, use the DW_SUPPRESS_WARN macro explained earlier.

Timing Waveforms

Figure 1-3 on page 8 shows a case where there are two pipeline register levels since *in_reg* = 0, *stages* = 2, and *out_reg* = 1. Launching is performed while *accept_n* is de-asserted causing the pipeline to fill up. This is indicated by *pipe_full* going to '1' while *accept_n* is '1'. The *pipe_census*[1:0] value is '2' which indicates that all the pipeline register levels contain active results.

At the point that the pipeline is full, *accept_n* is asserted ('0') to begin emptying the pipeline. Note that *pipe_full* de-asserts when *accept_n* is asserted, but the *pipe_census*[1:0] value still indicates '2' the next clock cycle since a launch coincided with the asserted *accept_n*. Once the launching activity ceases, the continued assertion of *accept_n* drains the pipeline of active quotient results with *pipe_census*[1:0] eventually going to '0'.

Figure 1-3 Launching Until Full, Accepting Until Empty

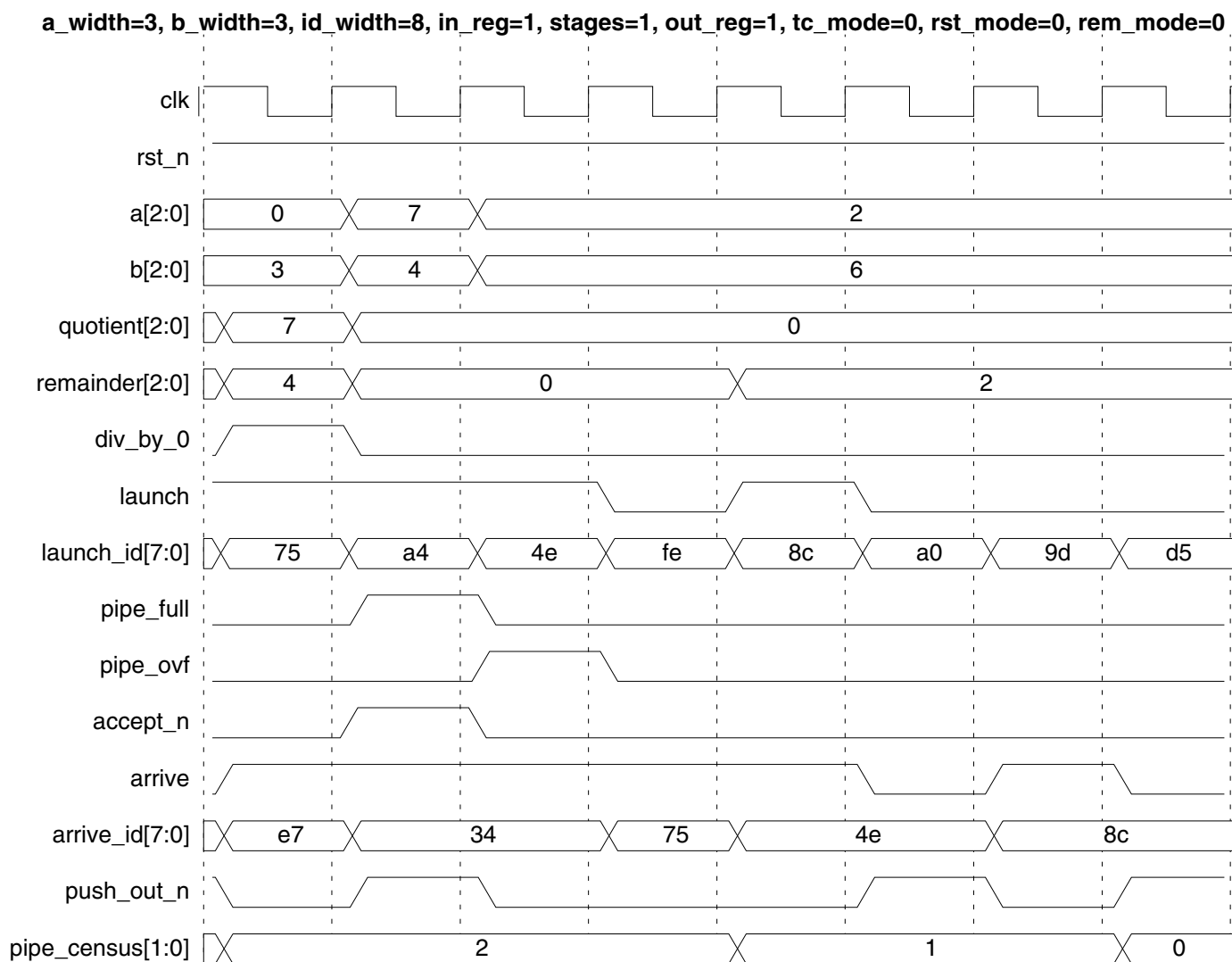


Figure 1-4 shows a case where `launch` is asserted every other clock cycle while `accept_n` is always asserted ('0'). There are 4 pipeline register levels. So, the quotient result of $221/59 = '3'$ (`launch_id[13:0]` of '0x236a') arrives after the fourth rising-edge of `clk`.

Figure 1-4 Launch Every Other Cycle with Asserted `accept_n`

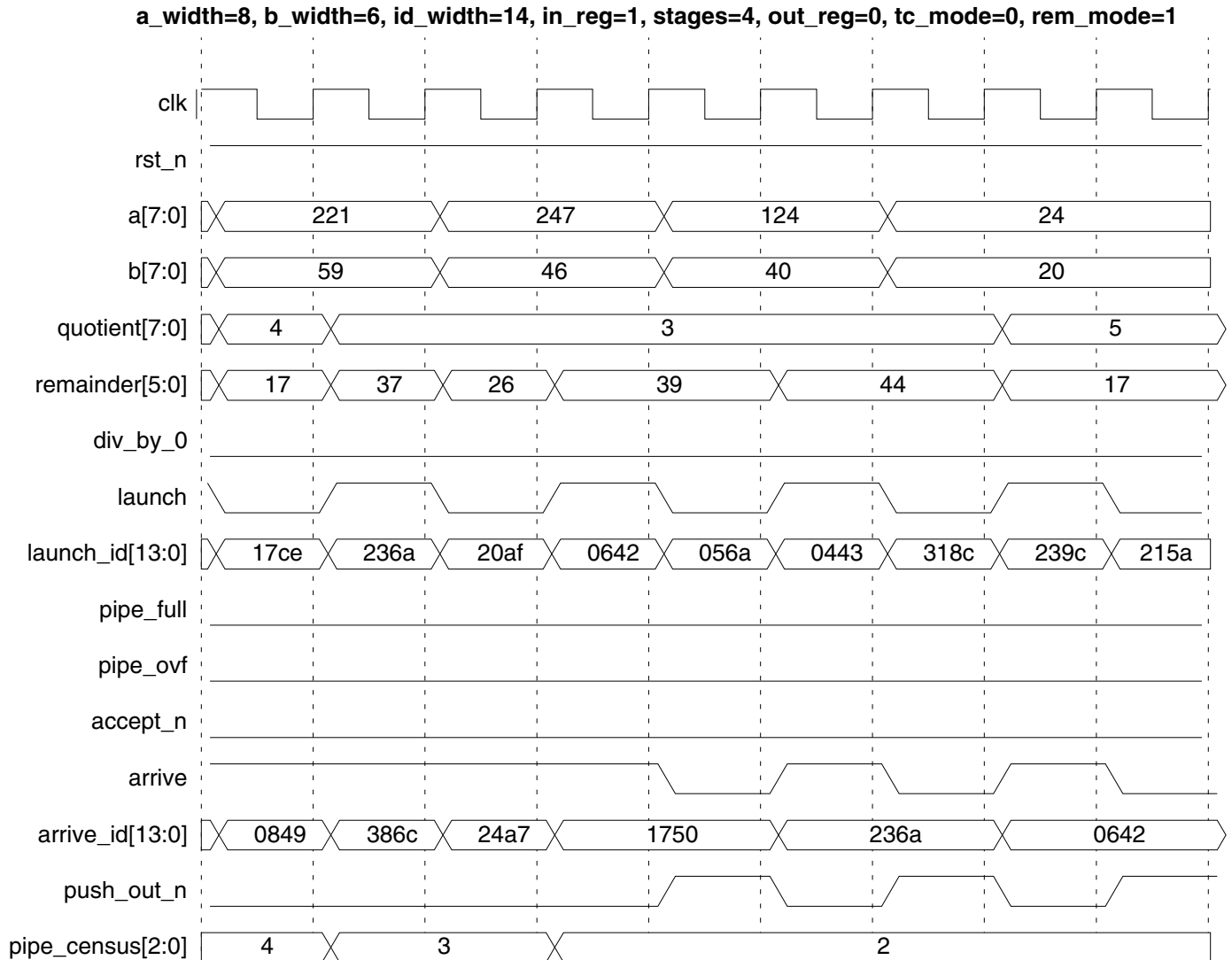


Figure 1-5 depicts a pipeline overflow condition. This is the same configuration as shown in Figure 1-4. The `pipe_ovf` output is registered and gets asserted following the rising-edge of `clk` when the pipeline is full (`pipe_full` is '1'), `launch` is asserted ('1'), and `accept_n` is not asserted ('1'). In this situation, the launched operation is ignored and the pipeline contents are preserved.

Figure 1-5 Pipeline Overflow

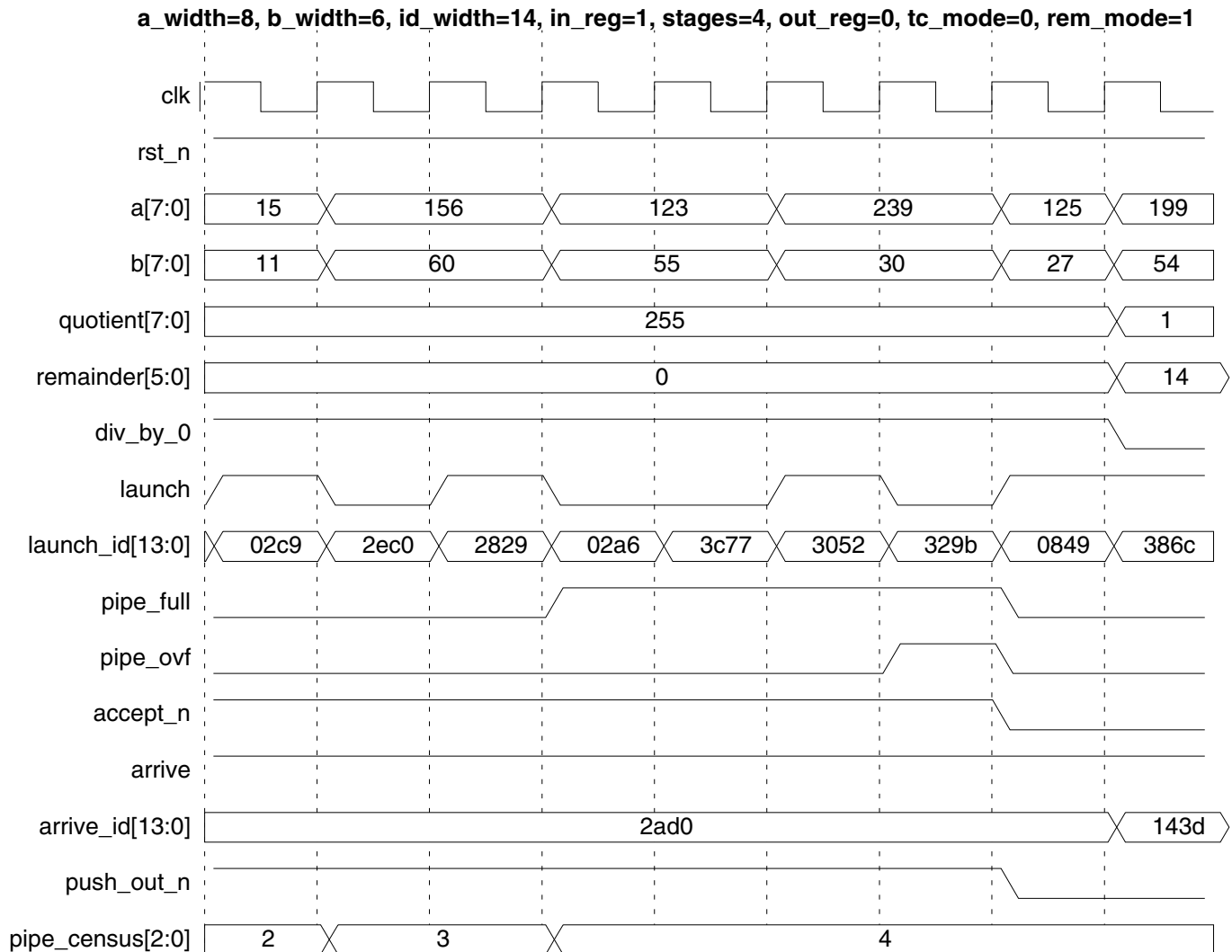


Figure 1-6 depicts a scenario when there is no pipelining configured into the DW_lp_piped_div, which is defined when $in_reg = 0$, $stages = 1$, and $out_reg = 0$. Thus, the quotient result is a pure combinational logic path from a and b. The flow control/status outputs arrive, arrive_id, pipe_full, pipe_ovf, push_out_n still have meaning. However, the output pipe_census has no meaning because no pipeline register levels exist. Hence, pipe_census is always driven to '0'.

Notice that when launch is asserted and accept_n is not, the register output pipe_ovf goes to '1'. This is due to the fact that when $accept_n = 1$, it implies that the downstream device cannot accept any more results. Thus, a launch under this condition results in overrun and the subsequent quotient is lost.

DW_lp_piped_div is configured to operate in two's complement mode.

Figure 1-6 No Pipeline Specified ($in_reg = 0$, $stages = 1$, $out_reg = 0$)

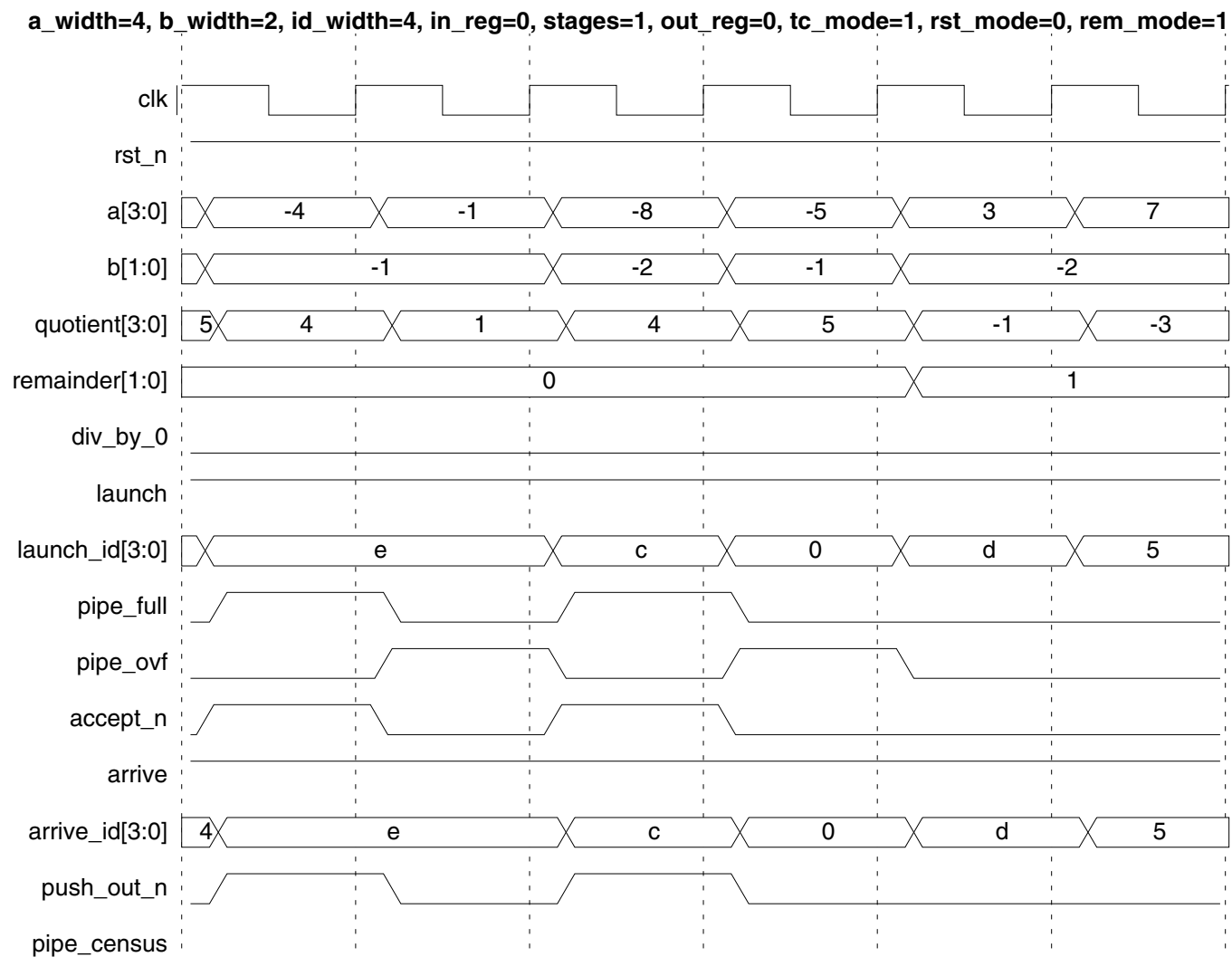
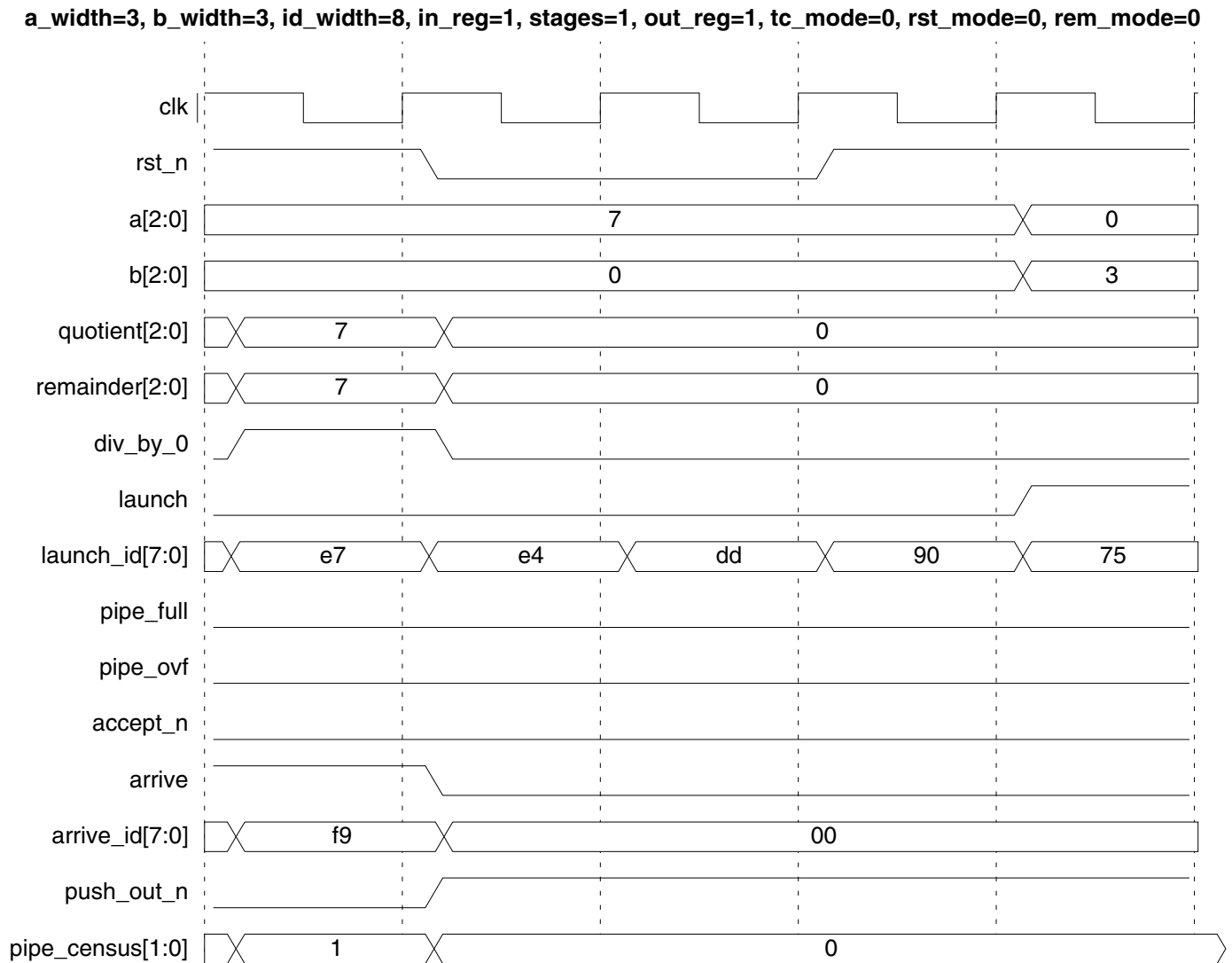


Figure 1-7 shows the affects of asserting of `rst_n` when configured for asynchronous resetting (`rst_mode=0`).

Figure 1-7 Asynchronous Reset Behavior (`rst_mode = 0`)



Enabling minPower

In Design Compiler (version P-2019.03 and later) and Fusion Compiler, you can instantiate this component and use all its features without special settings.

For versions of Design Compiler before P-2019.03, enable minPower as follows:

```
set synthetic_library {dw_foundation.sldb dw_minpower.sldb}
set link_library {* $target_library $synthetic_library}
```

Related Topics

- [DesignWare Building Blocks User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp.all;

entity DW_lp_piped_div_inst is
    generic (
        inst_a_width : POSITIVE := 8;
        inst_b_width : POSITIVE := 8;
        inst_id_width : POSITIVE := 8;
        inst_in_reg : NATURAL := 0;
        inst_stages : POSITIVE := 4;
        inst_out_reg : NATURAL := 0;
        inst_tc_mode : NATURAL := 0;
        inst_rst_mode : NATURAL := 0;
        inst_rem_mode : NATURAL := 1;
        inst_op_iso_mode : NATURAL := 0
    );
    port (
        inst_clk : in std_logic;
        inst_rst_n : in std_logic;
        inst_a : in std_logic_vector(inst_a_width-1 downto 0);
        inst_b : in std_logic_vector(inst_b_width-1 downto 0);
        quotient_inst : out std_logic_vector(inst_a_width-1 downto 0);
        remainder_inst : out std_logic_vector(inst_b_width-1 downto 0);
        div_by_0_inst : out std_logic;
        inst_launch : in std_logic;
        inst_launch_id : in std_logic_vector(inst_id_width-1 downto 0);
        pipe_full_inst : out std_logic;
        pipe_ovf_inst : out std_logic;
        inst_accept_n : in std_logic;
        arrive_inst : out std_logic;
        arrive_id_inst : out std_logic_vector(inst_id_width-1 downto 0);
        push_out_n_inst : out std_logic;
        pipe_census_inst : out
        std_logic_vector(bit_width(maximum(1,inst_in_reg+(inst_stages-1)+inst_out_reg)+1)-1
        downto 0)
    );
end DW_lp_piped_div_inst;

architecture inst of DW_lp_piped_div_inst is

begin

    -- Instance of DW_lp_piped_div
    U1 : DW_lp_piped_div

```

```
generic map ( a_width => inst_a_width,  
              b_width => inst_b_width,  
              id_width => inst_id_width,  
              in_reg => inst_in_reg,  
              stages => inst_stages,  
              out_reg => inst_out_reg,  
              tc_mode => inst_tc_mode,  
              rst_mode => inst_rst_mode,  
              rem_mode => inst_rem_mode,  
              op_iso_mode => inst_op_iso_mode )  
port map ( clk => inst_clk,  
           rst_n => inst_rst_n,  
           a => inst_a,  
           b => inst_b,  
           quotient => quotient_inst,  
           remainder => remainder_inst,  
           div_by_0 => div_by_0_inst,  
           launch => inst_launch,  
           launch_id => inst_launch_id,  
           pipe_full => pipe_full_inst,  
           pipe_ovf => pipe_ovf_inst,  
           accept_n => inst_accept_n,  
           arrive => arrive_inst,  
           arrive_id => arrive_id_inst,  
           push_out_n => push_out_n_inst,  
           pipe_census => pipe_census_inst );  
  
end inst;
```

HDL Usage Through Component Instantiation - Verilog

```

module DW_lp_piped_div_inst( inst_clk,
                             inst_rst_n,
                             inst_a,
                             inst_b,
                             quotient_inst,

                             remainder_inst,
                             div_by_0_inst,
                             inst_launch,
                             inst_launch_id,
                             pipe_full_inst,

                             pipe_ovf_inst,
                             inst_accept_n,
                             arrive_inst,
                             arrive_id_inst,
                             push_out_n_inst,

                             pipe_census_inst );

parameter a_width = 4;
parameter b_width = 2;
parameter id_width = 4;
parameter in_reg = 0;
parameter stages = 1;
parameter out_reg = 0;
parameter tc_mode = 1;
parameter rst_mode = 0;
parameter rem_mode = 1;
parameter op_iso_mode = 0;

`define m 1
`define bit_width_m 1

input inst_clk;
input inst_rst_n;
input [a_width-1 : 0] inst_a;
input [b_width-1 : 0] inst_b;
output [a_width-1 : 0] quotient_inst;
output [b_width-1 : 0] remainder_inst;
output div_by_0_inst;
input inst_launch;
input [id_width-1 : 0] inst_launch_id;
output pipe_full_inst;
output pipe_ovf_inst;
input inst_accept_n;

```

```
output arrive_inst;
output [id_width-1 : 0] arrive_id_inst;
output push_out_n_inst;
output [`bit_width_m-1 : 0] pipe_census_inst;

// Instance of DW_lp_piped_div
DW_lp_piped_div #(a_width,
                  b_width,
                  id_width,
                  in_reg,
                  stages,
                  out_reg,
                  tc_mode,
                  rst_mode,
                  rem_mode,
                  op_iso_mode)
U1 ( .clk(inst_clk),
     .rst_n(inst_rst_n),
     .a(inst_a),
     .b(inst_b),
     .quotient(quotient_inst),
     .remainder(remainder_inst),
     .div_by_0(div_by_0_inst),
     .launch(inst_launch),
     .launch_id(inst_launch_id),
     .pipe_full(pipe_full_inst),
     .pipe_ovf(pipe_ovf_inst),
     .accept_n(inst_accept_n),
     .arrive(arrive_inst),
     .arrive_id(arrive_id_inst),
     .push_out_n(push_out_n_inst),
     .pipe_census(pipe_census_inst) );

endmodule
```


Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	Updated date and version; no technical changes
December 2023	DWBB_202312.0	Updated date and version; no technical changes
July 2023	DWBB_202212.5	Updated date and version; no technical changes
July 2020	DWBB_201912.5	Adjusted content and title of “Suppressing Warning Messages During Verilog Simulation” on page 7 and added the DW_SUPPRESS_WARN macro and the divide-by-zero message
June 2020	DWBB_201912.4	- Corrected the description of <code>launch</code> in Table 1-1 on page 1 (typo) - Corrected the description of <code>rem_mode</code> in Table 1-2 on page 2 - Corrected the width of input <code>b</code> in the waveform in Figure 1-6 on page 11
October 2019	DWBB_201903.5	Added the “Disabling Clock Monitor Messages” section
March 2019	DWBB_201903.0	- Clarified the <code>op_iso_mode</code> parameter in Table 1-2 on page 2 - Clarified license requirements in Table 1-3 on page 3 - Added Figure 1-2 on page 5 to clarify datapath gating - Added “Enabling minPower” on page 12 - Added this Revision History table and the document links on this page

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